Introduction to the ER Model in DBMS

Entity, Attribute, Relationship, Degree, Cardinality, and Participation Constraint

Lecture Material

Course: Databases

Updated: 02/04/2026

Learning Objectives

After completing this lecture, learners will be able to:

1. Explain the concept of the **Entity-Relationship Model** in database design.
2. Describe the role of the **ER Model** and **ER Diagram**.
3. Distinguish the main components: **entity**, **attribute**, and **relationship**.
4. Identify **strong entities** and **weak entities**.
5. Distinguish attribute types: simple, key, composite, multivalued, and derived.
6. Explain **degree**, **cardinality**, and **participation constraint**.
7. Apply the basic process for drawing an ER Diagram for a practical problem.

Lecture Outline

1. Overview of the ER Model
2. Role in database design
3. Components of the ER Model
4. Entity, strong entity, and weak entity
5. Attribute and attribute types
6. Relationship and degree
7. Cardinality
8. Participation constraint
9. How to draw an ER Diagram
10. Examples, questions, and exercises

Overview of the ER Model

What is the ER Model?

The **Entity-Relationship Model**, or **ER Model**, is a conceptual model used to design databases.

This model represents the logical structure of a database through:

- The **entities** whose data must be stored.
- The **attributes** that describe those entities.
- The **relationships** among entities.

What Questions Does the ER Model Answer?

The ER Model helps designers answer:

- What objects does the system need to store?
- What information does each object have?
- How are the objects related to one another?
- What constraints should be represented in the data model?

ER Diagram

The graphical representation of the ER Model is called an **Entity-Relationship Diagram**, abbreviated as **ERD**.

Illustration: Overview of the ER Model

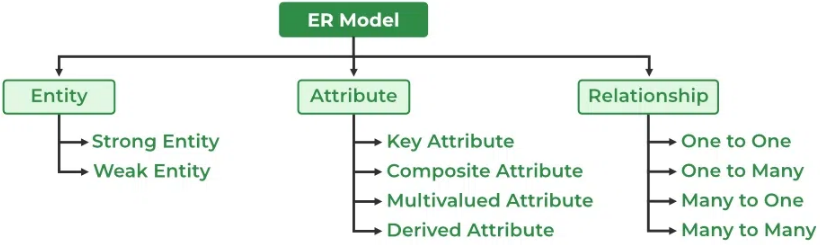


Figure 1. Overview of the ER Model.

Role of the ER Model in Database Design

The database design process usually involves three stages:

1. **Requirements gathering**: asking users, customers, or stakeholders about system functions and data requirements.
2. **Conceptual or logical design**: using the ER Model to describe entities, attributes, and relationships.
3. **Physical design**: defining tables, keys, indexes, views, and implementation details for a specific DBMS.

Why Are ER Diagrams Useful?

ER Diagrams are useful because they:

- Represent data and data relationships visually.
- Can be converted into relational tables.
- Model real-world objects.
- Do not require viewers to deeply understand a specific DBMS.
- Make complex systems easier to understand.

Quick Quiz 1: Overview of the ER Model

Question 1. What is the ER Model mainly used for?

A. Designing the conceptual model of a database
B. Increasing the CPU speed of a server
C. Designing user interfaces
D. Compressing image data

Question 2. What is an ER Diagram?

A. A data backup tool
B. A graphical representation of the ER Model
C. A database operating system
D. A web programming language

Quick Quiz 1: Overview of the ER Model (cont.)

Question 3. In which design step is the ER Model most commonly used?

A. Network card configuration
B. Source code compilation
C. Conceptual or logical design
D. Deleting old data

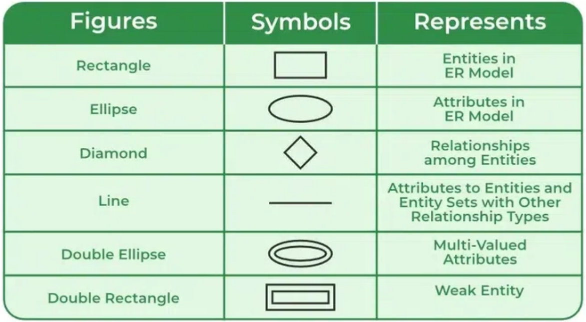
Main Components of the ER Model

Three Core Components

The ER Model consists of three core components:

Component	Meaning	Common ERD Symbol
Entity	An object or concept whose data must be stored	Rectangle
Attribute	A property that describes an entity	Oval
Relationship	A connection between entities	Diamond

Illustration: Components of an ER Diagram



Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Figure 2. Entity, Attribute, and Relationship in an ER Diagram.

Entity in the ER Model

What is an Entity?

An **Entity** is an object, thing, concept, or event in the real world about which the system needs to store information.

Examples:

- `Student`
- `Course`
- `Employee`
- `Company`
- `Product`
- `Reservation`
- `Device`
- `Document`

In a relational database, an entity is usually converted into a **table**.

Entity Type, Entity Instance, and Entity Set

Entity Type

Describes the general structure of one type of entity, for example: `Student(student_id, full_name, date_of_birth, email)`.

Entity Instance

A specific object belonging to an entity type, for example: `S001, Nguyen An, 2005-01-15, an@example.com`.

Entity Set

The collection of all entities of the same entity type, for example all students in a university.

Reminder about Entity Sets

An ER Diagram usually represents an **entity type** or an **entity set**, not each specific data row.

Example

An ERD represents the structure `Student`; it does not draw each specific student such as S001, S002, or S003.

Illustration: Entity Set

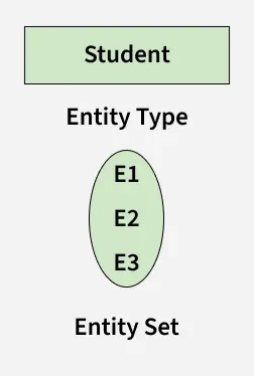


Figure 3. Entity type, entity instance, and entity set.

Strong Entity

A **Strong Entity** is an entity type that can be uniquely identified by its own attributes.

Characteristics:

- It has a **key attribute** or **primary key**.
- It does not depend on another entity for identification.
- It is usually represented by a **single rectangle** in an ER Diagram.

Example

`Student` has `student_id`; `Employee` has `employee_id`; `Product` has `product_id`.

Weak Entity

A **Weak Entity** is an entity type that cannot be uniquely identified by its own attributes alone.

A Weak Entity must depend on a **Strong Entity** for identification.

- It does not have a sufficiently strong key of its own.
- It depends on an identifying strong entity.
- It usually has **total participation** in its relationship with the strong entity.
- It is represented by a **double rectangle**.
- Its identifying relationship is usually represented by a **double diamond**.

Example of a Weak Entity

A company stores information about employees' dependents.

- `Employee` is a strong entity.
- `Dependent` is a weak entity.
- `Dependent` cannot exist independently without being linked to a specific employee.

Illustration: Strong Entity and Weak Entity

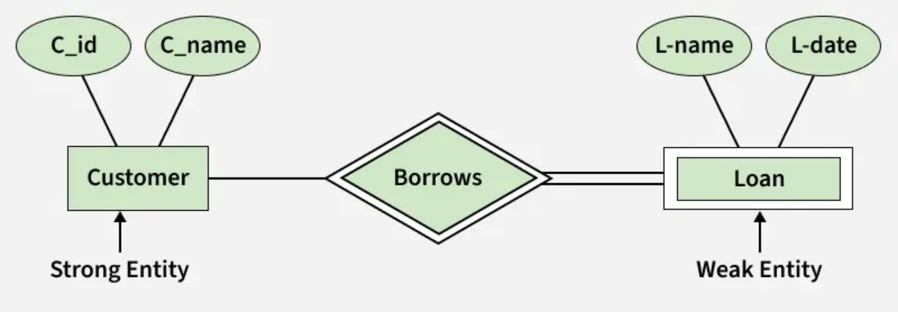


Figure 4. Strong Entity and Weak Entity.

Quick Quiz 2: Entity

Question 4. What is an entity in the ER Model?

A. A numeric data type
B. An index type
C. An SQL statement
D. A real-world object or concept whose data must be stored

Question 5. Which of the following is usually an example of a strong entity?

A. A secondary phone number with no owner
B. A temporary log row with no identifier
C. `Student` with a unique `student_id`
D. An attribute derived from date of birth

Quick Quiz 2: Entity (cont.)

Question 6. Which characteristic belongs to a Weak Entity?

A. It is always a multivalued attribute
B. It depends on a strong entity for identification
C. It cannot appear in an ER Diagram
D. It is always represented by a single oval

Attribute in the ER Model

What is an Attribute?

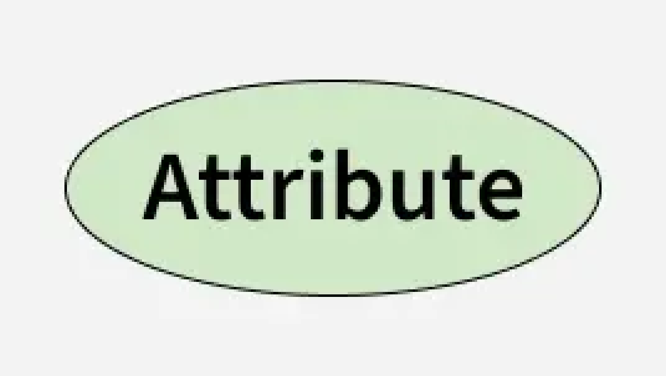
An **Attribute** is a property used to describe an entity.

For example, the entity `Student` may have attributes such as:

- `Roll_No`
- `Name`
- `DOB`
- `Age`
- `Address`
- `Mobile_No`

In an ER Diagram, an attribute is usually represented by an **oval**.

Illustration: Attribute



Types of Attributes

Common attribute types include:

1. **Simple Attribute**
2. **Key Attribute**
3. **Composite Attribute**
4. **Multivalued Attribute**
5. **Derived Attribute**

Simple Attribute

A **Simple Attribute** cannot be divided into smaller attributes with independent meaning.

Examples:

- `Age`
- `Gender`
- `Roll_No`

Note about the illustration

The original reference does not provide a separate illustration for Simple Attribute. In an ER Diagram, a simple attribute is still represented by a single oval, similar to ordinary attributes such as `Name`, `DOB`, or `Age`.

Key Attribute

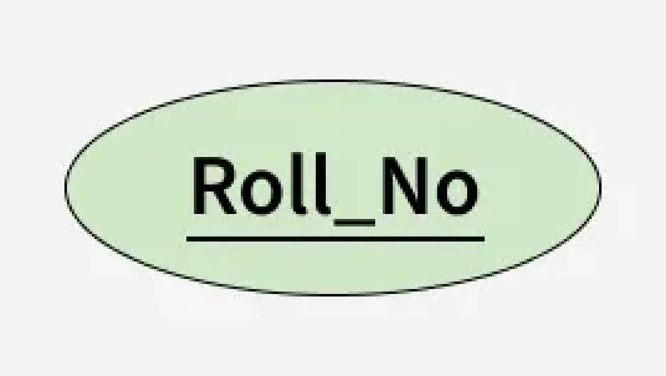
A **Key Attribute** is used to uniquely identify each entity in an entity set.

Examples:

- `Roll_No` uniquely identifies each student.
- `Employee_ID` uniquely identifies each employee.
- `Product_ID` uniquely identifies each product.

In an ER Diagram, a key attribute is represented by an underlined oval.

Illustration: Key Attribute



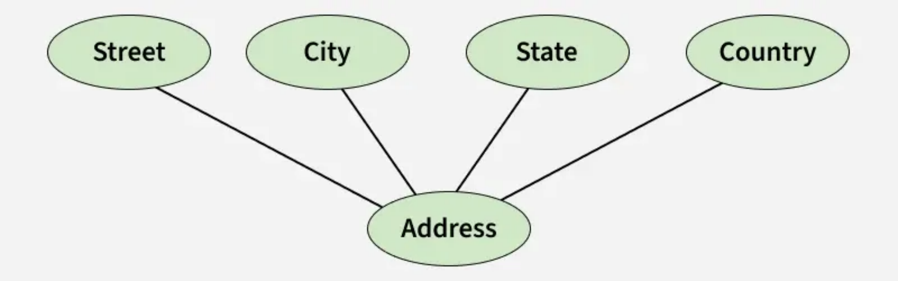
Composite Attribute

A **Composite Attribute** can be divided into smaller attributes.

For example, `Address` may consist of:

- `Street`
- `City`
- `State`
- `Country`

Illustration: Composite Attribute



Multivalued Attribute

A **Multivalued Attribute** can have more than one value for a single entity.

Examples:

- A student may have several phone numbers.
- An employee may have several skills.
- A product may have several colors.

In an ER Diagram, a multivalued attribute is represented by a double oval.

Illustration: Multivalued Attribute

Derived Attribute

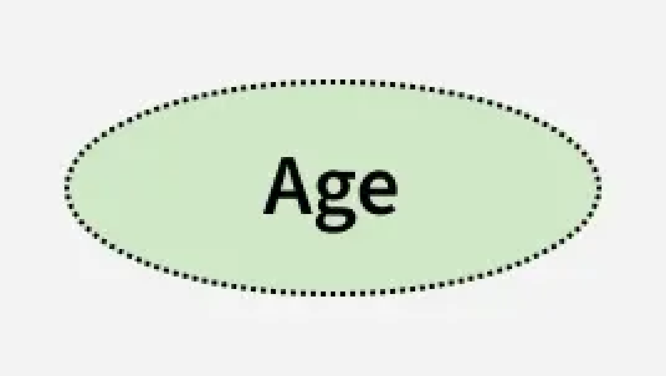
A **Derived Attribute** can be derived from another attribute.

Examples:

- Age can be derived from `DOB`.
- `Total_Amount` can be calculated from quantity and unit price.
- `Years_of_Service` can be calculated from the starting date.

In an ER Diagram, a derived attribute is represented by a dashed oval.

Illustration: Derived Attribute

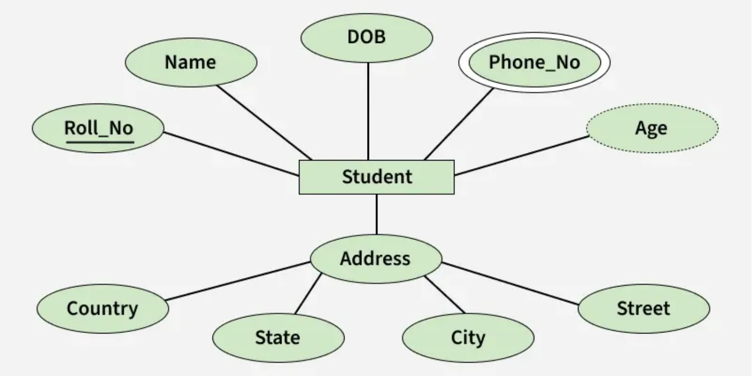


Entity and Combined Attributes

When attribute types are combined in one ER Diagram, the entity `Student` may have:

- `Roll_No` as a key attribute.
- `Name` and `DOB` as simple attributes.
- `Phone_No` as a multivalued attribute.
- `Age` as a derived attribute.
- `Address` as a composite attribute consisting of `Street`, `City`, `State`, and `Country`.

Illustration: Entity and Combined Attributes



Quick Quiz 3: Attribute

Question 7. What is an attribute used for in the ER Model?

A. Deleting the whole database
B. Increasing disk capacity
C. Describing a property of an entity
D. Replacing the operating system

Question 8. `Roll_No` uniquely identifies a student. What type of attribute is it?

A. Composite Attribute
B. Multivalued Attribute
C. Derived Attribute
D. Key Attribute

Quick Quiz 3: Attribute (cont.)

Question 9. `Address` consisting of `Street`, `City`, `State`, and `Country` is an example of which attribute type?

A. Composite Attribute
B. Derived Attribute
C. Multivalued Attribute
D. Weak Attribute

Question 10. `Age` calculated from `DOB` is an example of which attribute type?

A. Key Attribute
B. Derived Attribute
C. Composite Attribute
D. Multivalued Attribute

Relationship and Degree

Relationship in the ER Model

A **Relationship** describes a connection between entities.

Examples:

- `Student` **enrolls in** `Course`.
- `Employee` **works for** `Department`.
- `Customer` **places** `Order`.
- `Doctor` **performs** `Surgery`.

In an ER Diagram, a relationship is usually represented by a diamond and connected to entities by lines.

Relationship Type and Relationship Set

Relationship Type

Describes the type of association between entity types. Example: `Student -- Enrolled in -- Course`.

Relationship Set

The set of relationships of the same type, for example S1 enrolled in C2, S2 enrolled in C1, and S3 enrolled in C3.

Illustration: Relationship Type

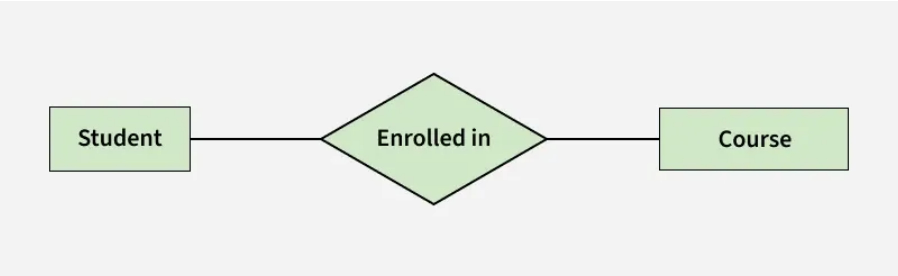


Figure 5a. Relationship type between Student and Course.

Illustration: Relationship Set

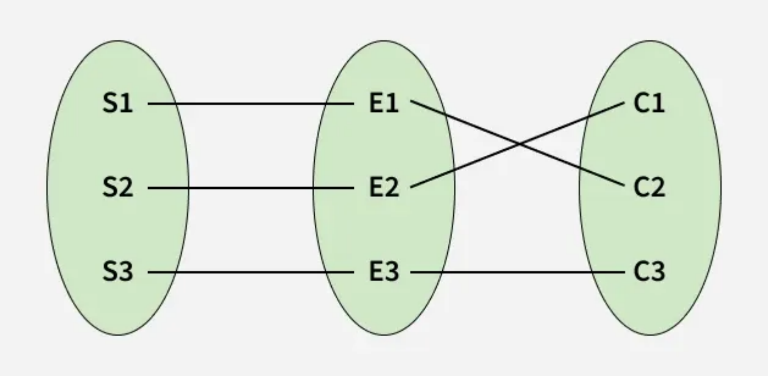


Figure 5b. Relationship set of the Enrolled in type.

Degree of a Relationship Set

The **Degree** of a relationship set is the number of different entity sets participating in the relationship set.

There are four common forms:

1. **Unary / Recursive Relationship**: one participating entity set.
2. **Binary Relationship**: two participating entity sets.
3. **Ternary Relationship**: three participating entity sets.
4. **N-ary Relationship**: n participating entity sets.

Examples of Degree

Degree Type	Example
Unary / Recursive	One employee manages another employee
Binary	A student enrolls in a course
Ternary	A supplier supplies a part to a project
N-ary	A relationship involving more than three entity sets

Illustration: Unary / Recursive Relationship

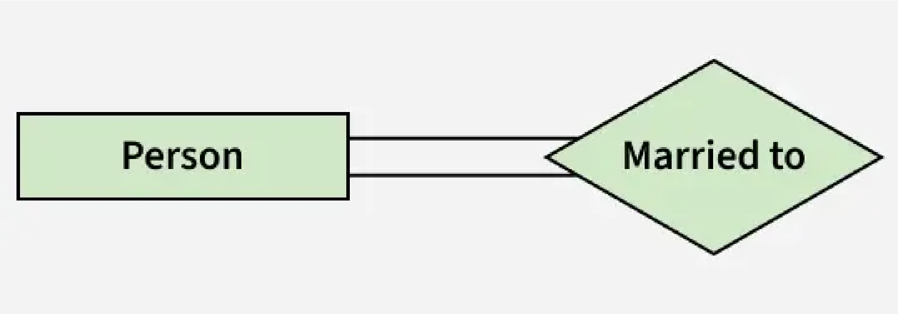


Illustration: Binary Relationship

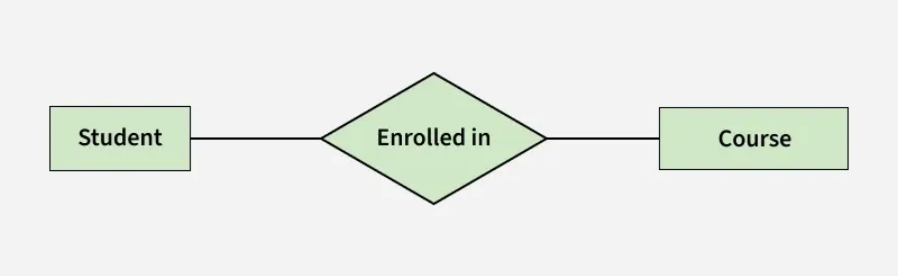


Illustration: Ternary Relationship

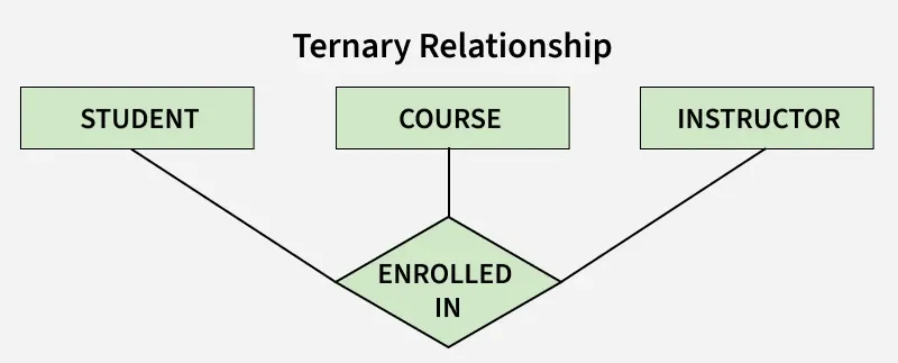
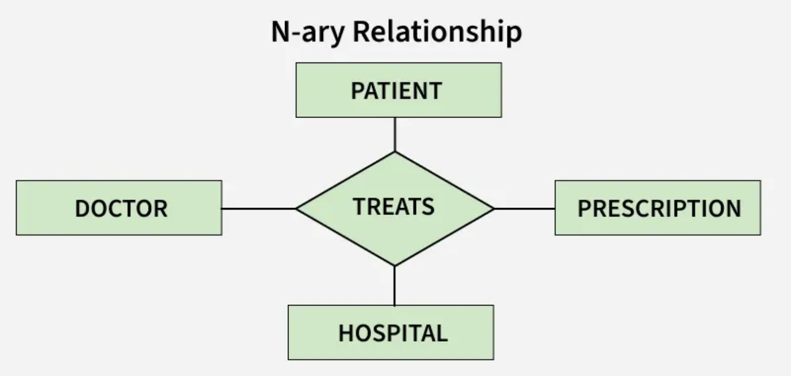


Illustration: N-ary Relationship



Quick Quiz 4: Relationship and Degree

Question 11. What does a relationship represent in the ER Model?

A. A font type
B. A connection between entities
C. A CPU performance metric
D. An image data type

Question 12. What is a relationship involving two entity sets called?

A. Unary Relationship
B. Ternary Relationship
C. Binary Relationship
D. N-ary Relationship

Quick Quiz 4: Relationship and Degree (cont.)

Question 13. A relationship in which one employee manages another employee is an example of which degree?

A. Ternary Relationship
B. Binary Relationship
C. N-ary Relationship
D. Unary / Recursive Relationship

Question 14. What does the degree of a relationship set indicate?

A. The number of entity sets participating in the relationship set
B. The number of records in the database
C. The number of columns in a table
D. The number of indexes created

Cardinality in the ER Model

What is Cardinality?

Cardinality indicates the maximum number of times an entity in an entity set can participate in a relationship set.

Cardinality helps answer:

- How many other entities can one entity be associated with?
- Is the relationship one-to-one, one-to-many, or many-to-many?
- When converting to relational tables, do we need foreign keys or an intermediate table?

One-to-One Relationship

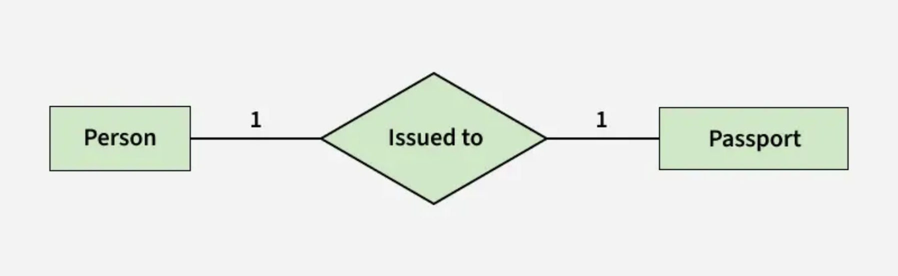
A **one-to-one** relationship occurs when each entity in each entity set participates at most once in the relationship.

Examples:

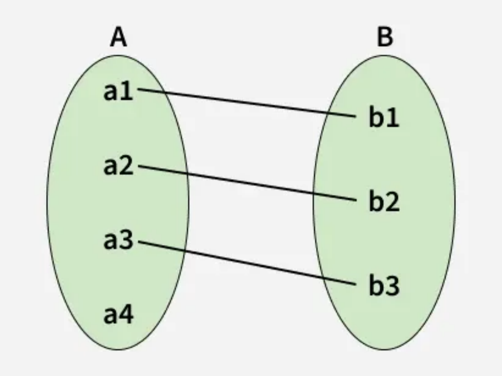
- One person has only one passport.
- One passport belongs to only one person.

```
Person 1 -- 1 Passport
```

Illustration: One-to-One



Set Representation: One-to-One



One-to-Many Relationship

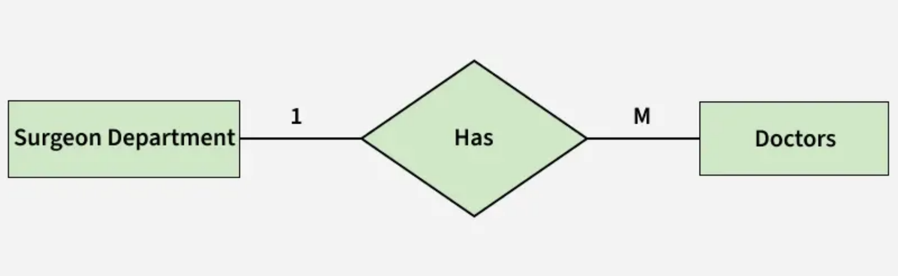
A **one-to-many** relationship occurs when one entity on the first side can be associated with multiple entities on the second side.

Examples:

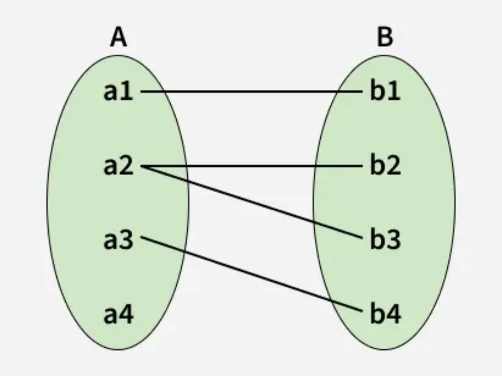
- One department has many doctors.
- One customer has many orders.
- One class has many students.

```
Department 1 -- M Doctor
```

Illustration: One-to-Many



Set Representation: One-to-Many



Many-to-One Relationship

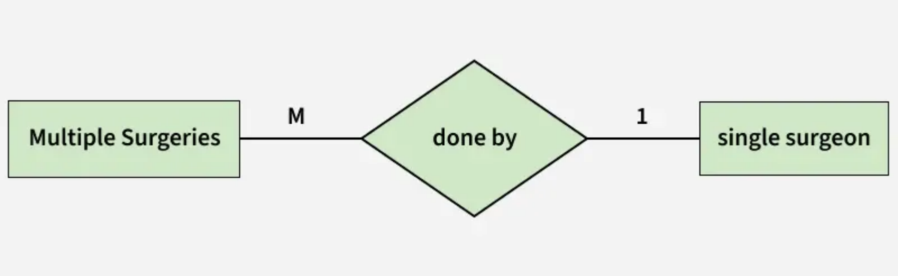
A **many-to-one** relationship is the reverse direction of one-to-many.

Examples:

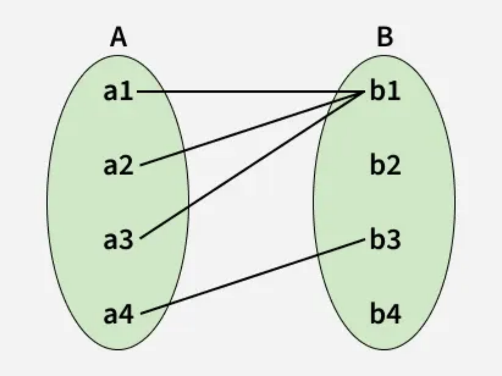
- Many surgeries can be performed by one surgeon.
- Many orders belong to one customer.
- Many employees belong to one department.

```
Surgery M -- 1 Surgeon
```

Illustration: Many-to-One



Set Representation: Many-to-One



Many-to-Many Relationship

A **many-to-many** relationship occurs when entities on both sides can participate multiple times in the relationship.

Examples:

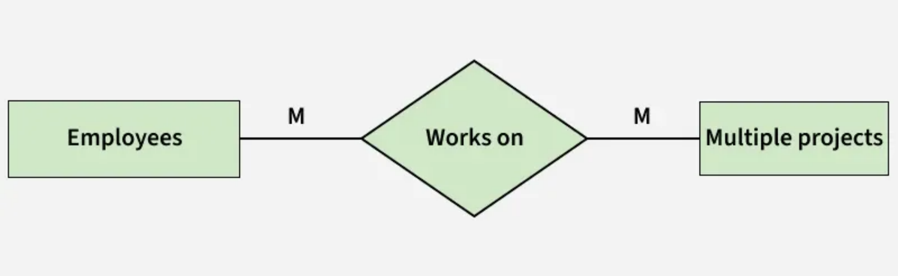
- A student studies many courses.
- A course has many students.
- An employee works on many projects.
- A project has many employees.

Intermediate Table for Many-to-Many

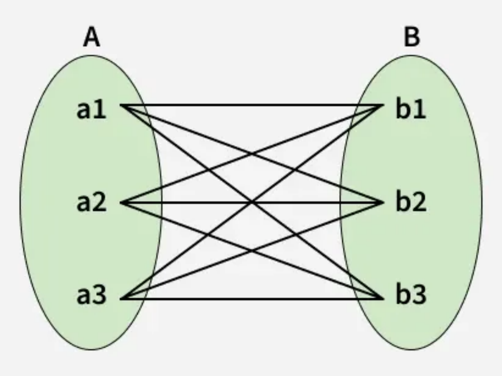
When converting to the relational model, a many-to-many relationship usually requires an intermediate table.

```
Students(student_id, full_name)
Courses(course_id, course_name)
Enrollments(student_id, course_id, semester, grade)
```

Illustration: Many-to-Many



Set Representation: Many-to-Many



Quick Quiz 5: Cardinality

Question 15. What does cardinality indicate in the ER Model?

A. The maximum number of times an entity can participate in a relationship set
B. The color of a rectangle in an ERD
C. The number of backup files
D. The SQL query speed

Question 16. One person has one passport, and one passport belongs to one person. What is this relationship?

A. One-to-Many
B. Many-to-One
C. One-to-One
D. Many-to-Many

Quick Quiz 5: Cardinality (cont.)

Question 17. One customer has many orders. What relationship type is this?

A. One-to-One
B. Many-to-Many
C. Many-to-One
D. One-to-Many

Question 18. If a student can take many courses and a course can have many students, what is the usual relationship type?

A. One-to-One
B. Many-to-Many
C. One-to-Many
D. Recursive

Participation Constraint

What is Participation Constraint?

A **Participation Constraint** describes whether an entity in an entity set is required to participate in a relationship.

There are two main types:

1. **Total Participation**
2. **Partial Participation**

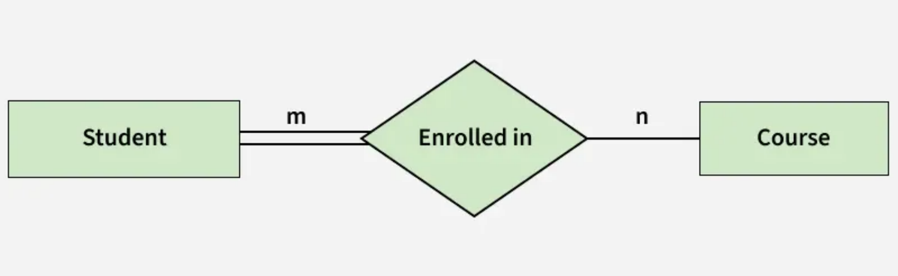
Total Participation

Total Participation means that every entity in the entity set must participate in the relationship.

Example: if the system requires every student to enroll in at least one course, then the entity set `Student` has total participation in the relationship `Enrolled in`.

In an ER Diagram, total participation is usually represented by a **double line**.

Illustration: Total Participation



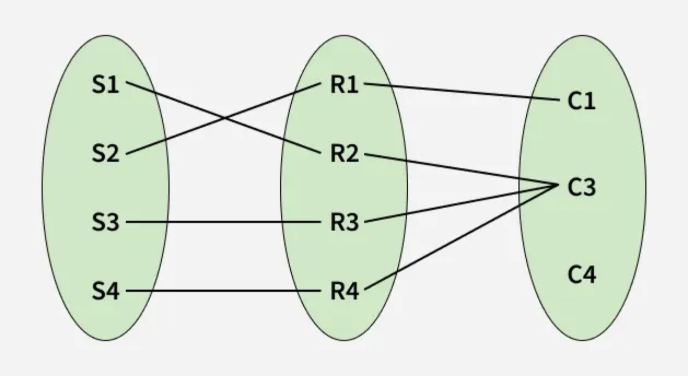
Partial Participation

Partial Participation means that an entity in the entity set may or may not participate in the relationship.

Example: some courses may not have any enrolled students yet. In that case, the entity set `Course` has partial participation in the relationship `Enrolled in`.

In an ER Diagram, partial participation is usually represented by a **single line**.

Set Representation: Participation Constraint



Quick Quiz 6: Participation Constraint

Question 19. What does Total Participation mean?

A. An entity never participates in a relationship
B. Only one entity participates in a relationship
C. All entities in the entity set must participate in the relationship
D. The relationship is removed from the ERD

Question 20. What does Partial Participation mean?

A. An entity may or may not participate in a relationship
B. An entity must participate in a relationship
C. An entity is always a weak entity
D. An entity always has many primary keys

How to Draw an ER Diagram

Process for Drawing an ER Diagram

An ER Diagram can be drawn in six steps:

1. Identify entities.
2. Identify relationships.
3. Add attributes.
4. Define primary keys.
5. Remove redundancies.
6. Review for clarity.

Step 1: Identify Entities

Identify all important entities in the system.

For example, in a student management system:

- `Student`
- `Course`
- `Lecturer`
- `Class`
- `Department`

Represent entities using rectangles.

Step 2: Identify Relationships

Identify relationships among entities.

Examples:

- `Student` enrolls in `Course`.
- `Lecturer` teaches `Class`.
- `Department` manages `Lecturer`.

Represent relationships using diamonds.

Step 3: Add Attributes

Attach attributes to entities using ovals.

For example, the entity `Student` may have:

- `student_id`
- `full_name`
- `date_of_birth`
- `email`
- `phone_number`

Step 4: Define Primary Keys

For each entity, identify the attribute that uniquely identifies it.

Examples:

- `Student`: `student_id`
- `Course`: `course_id`
- `Lecturer`: `lecturer_id`

In an ER Diagram, key attributes are usually underlined.

Steps 5 and 6: Finalize the ER Diagram

Step 5: Remove Redundancies

Remove unnecessary entities, duplicate relationships, repeated attributes, and relationships without business meaning.

Step 6: Review for Clarity

Check whether the diagram is easy to understand, can be converted into relational tables, and correctly reflects the system requirements.

Worked Example: Course Registration System

Example Problem

Consider a university course registration system.

We need to identify:

- Entities.
- Attributes.
- Relationships.
- Cardinalities.
- How to convert the ER model into relational tables.

Main Entities

Main entities:

- `Student`
- `Course`
- `Lecturer`
- `Department`

Attributes of the Entities

Student

- student_id
- full_name
- email
- date_of_birth

Course

- course_id
- course_name
- credits

Lecturer

- lecturer_id
- full_name
- email

Department

- department_id
- department_name

Main Relationships

Main relationships:

- `Student` **enrolls in** `Course`.
- `Lecturer` **teaches** `Course`.
- `Department` **offers** `Course`.
- `Department` **manages** `Lecturer`.

Cardinality in the Example

Some cardinalities:

- A student can enroll in many courses.
- A course can have many enrolled students.
- A lecturer can teach many courses.
- A course may be taught by one or more lecturers depending on the rules.
- A department can manage many lecturers.

Converting to Relational Tables

The ER model can be converted into the following tables:

```
Students(student_id, full_name, email, date_of_birth)
Courses(course_id, course_name, credits)
Lecturers(lecturer_id, full_name, email)
Departments(department_id, department_name)
Enrollments(student_id, course_id, semester, grade)
Teaches(lecturer_id, course_id, semester)
```

Interpreting Intermediate Tables

- `student_id` is the primary key of `Students`.
- `course_id` is the primary key of `Courses`.
- `lecturer_id` is the primary key of `Lecturers`.
- `department_id` is the primary key of `Departments`.
- `Enrollments` is an intermediate table for the many-to-many relationship between `Student` and `Course`.
- `Teaches` links `Lecturer` and `Course`.

Quick Quiz 7: Drawing an ERD

Question 21. When drawing an ER Diagram, what is usually the first step?

A. Choosing the background color of the diagram
B. Creating indexes for tables
C. Deleting all relationships
D. Identifying the important entities

Question 22. In an ER Diagram, what symbol usually represents a relationship?

A. Oval
B. Rectangle
C. Dashed line
D. Diamond

Questions and Exercises

Short-Answer Review Questions I

1. What is the ER Model? Why is it considered a conceptual model in database design?
2. Distinguish entity, attribute, and relationship.
3. Distinguish strong entity and weak entity. Give an example.
4. Present the common attribute types in the ER Model.
5. What is the degree of a relationship set? Give examples of unary, binary, and ternary relationships.
6. What is cardinality? Why is it important when converting an ER Model into a relational model?
7. Distinguish total participation and partial participation.

Exercise 1: University System

A university needs to build a system for managing students, courses, lecturers, and departments.

1. Identify the main entities.
2. List some attributes for each entity.
3. Identify relationships among the entities.
4. Determine the cardinality for each relationship.
5. Convert the ER model into a list of relational tables.

Exercise 2: Online Store

An online store needs to manage customers, products, orders, and payments.

1. Identify the main entities and attributes.
2. Identify relationships among `Customer`, `Order`, `Product`, and `Payment`.
3. Determine which relationships are one-to-many and which are many-to-many.
4. Propose intermediate tables if necessary.

Exercise 3: Employees and Dependents

A company wants to store information about employees and their dependents.

1. Identify the strong entity and weak entity.
2. Explain why one entity is a weak entity.
3. Identify the identifying relationship.
4. Suggest how to convert the model into relational tables.

Exercise 4: Hospital System

A hospital needs to manage patients, doctors, departments, surgeries, and medicines.

1. Identify at least five entities.
2. Identify at least five relationships.
3. Classify some relationships by degree and cardinality.
4. Indicate which relationships may require intermediate tables.

Summary and Keywords

Lesson Summary

- The ER Model is a conceptual model used for database design.
- An ER Diagram is the graphical representation of the ER Model.
- The three main components of the ER Model are entity, attribute, and relationship.
- An entity is an object or concept whose data must be stored.
- A strong entity has its own identifier; a weak entity depends on a strong entity.
- Attributes describe entities and include simple, key, composite, multivalued, and derived attributes.

Lesson Summary (cont.)

- Relationships represent connections between entities.
- Degree indicates the number of entity sets participating in a relationship.
- Cardinality indicates the maximum number of times an entity can participate in a relationship.
- Participation constraints indicate whether participation is mandatory or optional.
- The ERD drawing process includes identifying entities, relationships, attributes, primary keys, removing redundancies, and reviewing clarity.

Key Terms

- Entity-Relationship Model
- ER Model
- Entity-Relationship Diagram
- ERD
- Entity
- Entity Type
- Entity Set
- Entity Instance
- Strong Entity
- Weak Entity
- Attribute
- Simple Attribute
- Key Attribute
- Composite Attribute
- Multivalued Attribute
- Derived Attribute
- Relationship
- Relationship Type
- Relationship Set
- Degree
- Unary Relationship
- Binary Relationship
- Ternary Relationship
- N-ary Relationship
- Cardinality
- One-to-One
- One-to-Many
- Many-to-Many
- Participation Constraint
- Total/Partial Participation

Thank you!

Questions and Discussion